Yuhao Bao

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RESEARCH INTERESTS

Human-centered AI reliability and evaluation; how vision-language and large language models interpret crisis-related social media, and how their outputs compare with human annotations.

EDUCATION

Brigham Young University, Provo, UT — Expected April 2027
B.S. in Computer Science; Minors in Mathematics and Physics
GPA: 3.85/4.00 (Computer Science: 3.90/4.00)
Relevant coursework: Advanced Deep Learning; Multi-Agent Systems; Database Modeling

PUBLICATION

Bao, Y., & Hughes, A. (2026). When Social Media Images Need Words: Measuring Context Gap and Fusion Tax in Crisis Image Captioning. *Proceedings of the International ISCRAM Conference, 23*. Work-in-Progress paper. doi:10.59297/4bar0c32

RESEARCH EXPERIENCE

Research Assistant, HCI Research Group, Brigham Young University — Nov 2023 – Present
Advisor: Prof. Amanda Hughes
Crisis image captioning with vision-language models (ISCRAM 2026)

- Compared three vision-language models (Gemini 2.0 Flash, Qwen2.5-VL, BLIP) on 204 CrisisFACTS image-post pairs under image-only and image-plus-text conditions.
- Designed the scoring rubric and error taxonomy and carried out the human coding (Krippendorff's $\alpha = 0.69$).
- Built Python batch evaluation pipelines with JSON parsing and rubric-based scoring; adding post text raised Gemini captioning accuracy from 75.5% to 91.7% and reduced object misidentification from 15.7% to 2.9%.

Research Project: Text-to-SQL with Retrieval and LoRA — Fall 2025 – Winter 2026
Brigham Young University. Advisor: Prof. Porter Jenkins

- Built a Text-to-SQL RAG pipeline on Qwen3.5-9B with BM25 top-5 retrieval, fine-tuned on 6,169 examples and validated on 1,089.
- Implemented Shadow-LoRA (Base-to-Instruct adapter transfer) against a matched Direct-LoRA baseline; exact match rose from 19.47% to 30.49% and LLM-judge pass rate from 56.03% to 58.53%. Analyzed a reproducibility gap against the historical best (37.28% exact match).

INDUSTRY EXPERIENCE

Software Engineering Intern, Jingus (J.ing), Guangzhou, China — May 2023 – Aug 2023

- Developed Shopify Liquid theme components for an apparel storefront and integrated the PayPal payment gateway via REST API and webhooks to automate order status sync.

PROJECT

FitStack Macro Tracker (GitHub): nutrition tracker built with Streamlit, PostgreSQL (Supabase), and Plotly, with a normalized relational schema and cache-aside reads.

SKILLS

Research methods: rubric and error-taxonomy design, qualitative coding, inter-rater reliability, LLM and VLM evaluation
Programming: Python, SQL, Java, C++, JavaScript/TypeScript
ML and data: PyTorch, Hugging Face Transformers, LoRA/PEFT, BM25 retrieval, pandas, NumPy, PostgreSQL, Git, Linux

HONORS AND AFFILIATIONS

BYU travel funding to attend the 2024 SHPE National Convention (selected, limited awards)
Member: SHPE, ACM, BYU AI Club